

A Unified Representation Model and View Mapping Algorithm for Path-Mapping-Based Redirection Gains in Redirected Walking

Xiyu Bao
Shandong University
Jinan, Shandong, China
201820489@mail.sdu.edu.cn

Yulong Bian^{*†}
Shandong University
Weihai, China
bianyulong@sdu.edu.cn

Shiqing Xin
Shandong University
Jinan, China
xinshiqing@sdu.edu.cn

Meng Qi^{*}
School of Information Science and
Engineering
Shandong Normal University
Jinan, China
qimeng@sdnu.edu.cn

Chenglei Yang
School of Computer Science &
Technology
Shandong University
Jinan, China
chl_yang@sdu.edu.cn
bianyulong@sdu.edu.cn

Abstract

Among various redirected walking techniques, redirection gains guide users' visual perception by introducing deviations between virtual and physical walking trajectories. This allows users wearing virtual reality headsets to explore a larger virtual space within a limited physical area. Current redirection gain manipulation methods primarily establish mapping relationships between virtual and physical paths, typically represented by straight lines or arcs, to align users' physical and virtual perspectives. In this study, we propose a unified representation model and viewpoint mapping algorithm for path-mapping-based redirection gains. By inputting a few parameters, the model generates corresponding path mappings for various redirection gains, significantly simplifying the design and application of redirection gains. Furthermore, our viewpoint mapping algorithm effectively addresses mapping errors caused by dynamic gains in existing algorithms, which often lead to discontinuous trajectories and teleportation artifacts. This algorithm improves the continuity of the mapped trajectories and enhances the overall user experience during virtual exploration.

CCS Concepts

• Human-centered computing → Virtual reality.

Keywords

Redirected walking, Redirection gain, viewpoint mapping, representation model

^{*}Corresponding author

[†]The second affiliation is Shandong Key Laboratory of Intelligent Electronic Packaging Testing and Application, Weihai, China

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1 Introduction

Since redirected walking (RDW) technology [15] enables users to explore a larger virtual space by physical walking within a limited physical area, it has become a key interaction method in VR systems based on headsets in recent years. Among the various RDW techniques, redirection gains manipulate users' visual perception by introducing deviations between virtual and physical walking trajectories. These redirection gains, including translation [15], curvature [15], bending [9], gradient gain [13], and slope gain [5], each require establishing a mapping relationship between virtual and physical paths, which are typically represented as straight lines or arcs. This mapping allows the virtual (or physical) viewpoint to be determined based on the user's position along the physical (or virtual) path.

In redirection gains, gain values quantify the degree of deviation between virtual and physical paths. Early VR applications utilizing redirection gains primarily employed constant gain values during navigation. More recently, studies have introduced more flexible designs for redirection controllers, either by combining different gain methods [8] or applying specific gain methods with varying values across different paths [14, 18]. Despite these advances, gain values typically remain constant within each path, and sudden changes in gain values between adjacent paths can cause discomfort to users [1, 11]. To address this issue, dynamic gain methods have recently been proposed [2, 16], which dynamically adjust gain values to prevent sudden changes from being noticeable. Notably, in dynamic curvature gain, the virtual path is represented as a straight line, while the physical path follows a Clothoid curve. In contrast, dynamic bending gain uses a Clothoid curve for the virtual path and an arc for the physical path. Although the paths in dynamic gain methods differ from those in earlier approaches, the

fundamental principle of path mapping remains the same. When a user strictly follows the physical path, the viewpoint mapping is accurate, allowing the user to continuously move along the virtual path in the virtual environment. In practice, however, users often deviate from the path. The degree of this deviation affects the mapping calculation, leading to issues such as non-uniqueness in the mapping results when handling dynamic gains with existing methods. This can cause disruptions, such as teleportation, that interfere with the user's roaming experience.

This paper aims to explore a robust viewpoint mapping method to improve the user experience during redirection walking. This is particularly challenging for developers as it involves creating distinct viewpoint mapping strategies for each type of redirection gain. To address this problem, we first analyze and extract the common features of path-mapping-based gains and propose a unified representation model. This model describes the virtual and physical paths, as well as their mapping relationships under different redirection gain methods, within a general framework using a 5-tuple. Based on this model, we design a unified and robust viewpoint mapping algorithm that resolves mapping errors that occur when applying existing mapping algorithms in dynamic gain scenarios, ensuring the continuity of the mapping trajectory. In doing so, it not only ensures a good user experience when exploring larger virtual spaces within a confined physical environment, but also significantly simplifies the design and application of redirection gains, providing new insights and solutions for the further development of RDW technologies.

2 Related Works

RDW techniques involve various redirection gain methods, including translation gain, rotation gain, curvature gain [15], bending gain [9], vertical gain [12], gradient gain [13], and slope gain [5]. Among these, rotation gain modifies the relationship between virtual and physical rotations, and vertical gain maps the user's physical vertical movement into virtual space, with both gains not involving the mapping between physical and virtual paths. In contrast, the other gains require calculating the mapping relationship between the virtual and physical paths. Specifically, translation gain scales the user's forward movement in virtual space, mapping the virtual path to a longer/shorter physical path. Gradient gain introduces an inclination angle between the virtual walking plane and the physical ground, allowing users to perceive walking uphill or downhill in virtual space while walking on a flat plane. Unlike gradient gain, the virtual space in slope gain is not a flat plane. Curvature gain alters the curvature of walking path, enabling users to walk along a curved physical path without noticing, while perceiving themselves walking in a straight line in the virtual space. Similarly, bending gain modifies the walking path curvature, allowing users to walk along a curved physical path while perceiving themselves walking along a virtual path with a larger/smaller radius.

In redirection gains, determining the appropriate gain values is a key area of research. To ensure that redirection gain methods do not disrupt the sense of immersion, gain values must be set below a detection threshold (DT) to prevent users from noticing inconsistencies between the virtual and physical paths. Existing research focuses mainly on exploring various gain methods and testing their

corresponding DTs [3]. Some studies have also introduced factors such as gender, walking speed, and passive haptic cues to examine their effects on DT [4, 6, 7, 10].

However, most existing methods are based on constant gains. In some applications, redirection walking manipulations often combine various gain methods. To achieve smooth transitions between different gain values and reduce user discomfort, dynamic gain methods have been proposed [2, 16]. Sakono et al. [16] introduced dynamic curvature gain and dynamic bending gain methods, which vary the path curvature linearly with walking distance. These methods effectively reduce user discomfort. In addition, Sakono et al. also presented a viewpoint mapping approach, which maps the user's physical position to the closest point on the physical path. Based on the relationship between the physical and virtual paths, the corresponding point on the virtual path is then determined. Finally, the virtual position of the user is derived from the physical position offset, and the orientation of the user is processed similarly. This method is also applicable to other constant gain methods. However, in our practical application and experiments, we found that this viewpoint mapping method encounters issues when dealing with dynamic gains, such as non-unique mapping results. This leads to discontinuous trajectories after mapping, negatively impacting the user's roaming experience.

3 Methodology

In this section, we propose a unified representation model designed to describe virtual and physical paths and their mapping relationships under different redirection gain methods within a general framework. Based on this model, we then present a viewpoint mapping algorithm for redirection gain scenarios, aiming to ensure that when users deviate from the predefined physical (or virtual) path, their virtual (or physical) viewpoint remains continuous. We analyze existing redirection gain methods based on path-mapping, including translation gain ($g_T = T_v/T_p$, where T_v and T_p represent the virtual and physical walking distances, respectively), curvature gain ($g_C = C_p = 1/r_p$, where C_p and r_p represent the curvature and radius of the physical path), bending gain ($g_B = C_v/C_p = r_p/r_v$, where C_v and r_v represent the curvature and radius of the virtual path), and gradient gain ($g_G = T_h/T_p$, where T_h represents the virtual height), along with combinations of translation and curvature gains. We begin by analyzing the characteristics of these path-mapping-based gains. Since previous work has rarely described viewpoint mapping algorithms specific to these gain methods, we summarize the corresponding viewpoint mapping algorithms for each of these methods.

3.1 Unified Representation Model for Path-Mapping-Based Redirection Gains

Note: This is a note for the table.

The redirection gains mentioned above involve the following elements: the curvature of the physical path (C_p), the curvature of the virtual path (C_v), the distance ratio between the virtual and physical paths (T), the slope of the physical path (i_p), and the slope of the virtual path (i_v). For different redirection gains, the values of these elements vary. For example, in the case of translation gain, the corresponding values are: $C_p(s) = 0$, $C_v(s) = 0$, $T(s) = g_T$,

Table 1: Relationship between Unified Representation Model and Existing Redirection Gains

Gains	$c_v(s)$	$c_p(s)$	$T(s)$	$i_v(s)$	$i_p(s)$
Constant translation gain [15]	0	0	$g_T * s$	0	0
Constant curvature gain [15]	0	g_C	s	0	0
Constant bending gain [9]	r_p/g_B	$1/r_p$	s	0	0
Constant gradient gain [13]	0	0	s	$g_G * s$	0
Constant slope gain [5]	0	0	s	i_v	i_p
Linear translation gain [17]	0	0	$b_T * s^2$	0	0
Linear curvature gain [16]	0	$b_C * s$	s	0	0
Linear bending gain [16]	$1/r_p + b_B * s$	$1/r_p$	s	0	0
Quadratic curvature gain	0	$a_C * s^2$	s	0	0
Quadratic bending gain	$1/r_p + a_B * s^2$	$1/r_p$	s	0	0
Combined gain [8]	0	g_C	$G_T * s$	0	0

g_T , g_C , and g_B are the constant gain values for corresponding gains, respectively.

a_C and a_B are the second-order change rates for corresponding gains.

b_T , b_C , and b_B are the linear gain change rates for corresponding gains. r_p is the radius of the physical path.

$i_p(s) = 0$, and $i_v(s) = 0$. In contrast, for the curvature gain, the values are: $C_p(s) = g_C$, $C_v(s) = 0$, $T(s) = s$, $i_p(s) = 0$, and $i_v(s) = 0$. Based on these characteristics, we propose a unified representation model, the details of which will be introduced later.

We first present a unified path-mapping-based redirection gain representation model. Based on the above analysis, we show that by defining the 5-tuples ($C_p(s)$, $C_v(s)$, $T(s)$, $i_p(s)$, $i_v(s)$), we can represent various types of redirection gain. For example, curvature manipulation can be described using only C_v and C_p , while the translation gain is represented by T alone. Gradient gain requires both i_p and i_v . For combined gains, the corresponding elements are integrated according to the specific gain method. Table 1 shows the values of the corresponding elements for each gain type.

Within the framework of the unified gain representation model, both physical and virtual paths can be mathematically characterized using three-dimensional parametric representations. The physical path $P_p(s)$ and virtual path $P_v(s)$ are defined by Equation 1. $P_p(s)$ and $P_v(s)$ represent the spatial coordinates at a distance s from the path origin. The physical path coordinates ($X_p(s)$, $Y_p(s)$, $Z_p(s)$) and virtual path coordinates ($X_v(s)$, $Y_v(s)$, $Z_v(s)$) are explicitly expressed through Equation 2.

$$P_p(s) = (X_p(s), Y_p(s), Z_p(s)), P_v(s) = (X_v(s), Y_v(s), Z_v(s)) \quad (1)$$

The tangent direction of the physical path $v_p(s)$ and the virtual path $v_v(s)$ are defined by Equation 3. $v_p(s)$ and $v_v(s)$ represents the tangent direction of the position at a distance s from the path origin. The physical direction ($X_p(s)$, $Y_p(s)$, $Z_p(s)$) and virtual direction ($X_v(s)$, $Y_v(s)$, $Z_v(s)$) are explicitly expressed through Equations 4.

$$v_p(s) = (q1_p(s), q2_p(s), q3_p(s)), v_v(s) = (q1_v(s), q2_v(s), q3_v(s)) \quad (3)$$

Herein, $T(s)$ represents the distance ratio between them, $i_p(s)$ and $i_v(s)$ represent the slopes. This model provides a unified framework to represent all of the above path-mapping-based gains, including their dynamic variants.

3.2 Proposed Viewpoint Mapping Algorithm

In path-mapping-based gains, the existing viewpoint mapping method finds the closest perpendicular point on the physical path relative to the user's viewpoint and determines the corresponding point on the virtual path. The virtual viewpoint is then calculated based on the deviation between the physical viewpoint and its corresponding point. This process can also be reversed. This paper focuses on mapping from the physical to the virtual viewpoint, where the physical viewpoint consists of position p_p and orientation u_p , and the virtual viewpoint includes position p_v and orientation u_v . The mapping process for constant gains is shown in Fig. 1(a) and Fig. 1(b).

Current research on dynamic gains includes linear translation gain, linear curvature gain, and linear bending gain, where the gain values change linearly. The linear translation gain [17] modifies the mapping between q_p and q_v , while keeping both paths straight, so the existing algorithm remains valid. In Sakono et al.'s dynamic curvature gain [16], the virtual path is a straight line, while the physical path follows a Clothoid curve with a linearly changing curvature. For dynamic bending gain [16], the virtual path is a Clothoid curve and the physical path is a circular arc, as shown in Fig. 1(c). The formula for the Clothoid curve is given by: $X = X_0 + \int_0^s \cos\theta ds$, $Y = Y_0 + \int_0^s \sin\theta ds$, $\theta = c_0 + c_1s + c_2s^2/2$.

Sakono et al.'s mapping algorithm performs well with constant gains, providing accurate viewpoint mapping. However, in dynamic gain scenarios, when users deviate from the path, issues such as non-unique or incorrect results emerge, especially with larger deviations. For example, with dynamic bending gain, the Clothoid curve can have two nearest points for a given point p in virtual space (Fig. 1(c)), causing the physical position to alternate between q_1 and q_2 . In another scenario, as shown in Fig. 1(d), the nearest point on the Clothoid curve may switch from q_1 to q_2 , leading to oscillations in the user's physical position between p'_1 and p'_2 . A similar issue arises with dynamic curvature gain, where the virtual position may switch between points, causing oscillations in the user's physical

$$\begin{cases} X_p(s) = X_p(0) + \int_0^s \cos\left(v_p(0) + \int_0^s c_p(t) dt\right) ds \\ Y_p(s) = Y_p(0) + \int_0^s \sin\left(v_p(0) + \int_0^s c_p(t) dt\right) ds \\ Z_p(s) = Z_p(0) + i_p(s) \end{cases} \quad \begin{cases} X_v(s) = X_v(0) + \int_0^s \cos\left(v_v(0) + \int_0^s c_v(T^{-1}(t)) dt\right) ds \\ Y_v(s) = Y_v(0) + \int_0^s \sin\left(v_v(0) + \int_0^s c_v(T^{-1}(t)) dt\right) ds \\ Z_v(s) = Z_v(0) + i_v(s) \end{cases} \quad (2)$$

$$\begin{cases} q1_p(s) = \cos\left(\theta_p(0) + \int_0^s c_p(t) dt\right) \\ q2_p(s) = \sin\left(\theta_p(0) + \int_0^s c_p(t) dt\right) \\ q3_p(s) = i'_p(s) \end{cases} \quad \begin{cases} q1_v(s) = \cos\left(v_v(0) + \int_0^s c_v(T^{-1}(t)) dt\right) \\ q2_v(s) = \sin\left(v_v(0) + \int_0^s c_v(T^{-1}(t)) dt\right) \\ q3_v(s) = i'_v(s) \end{cases} \quad (4)$$

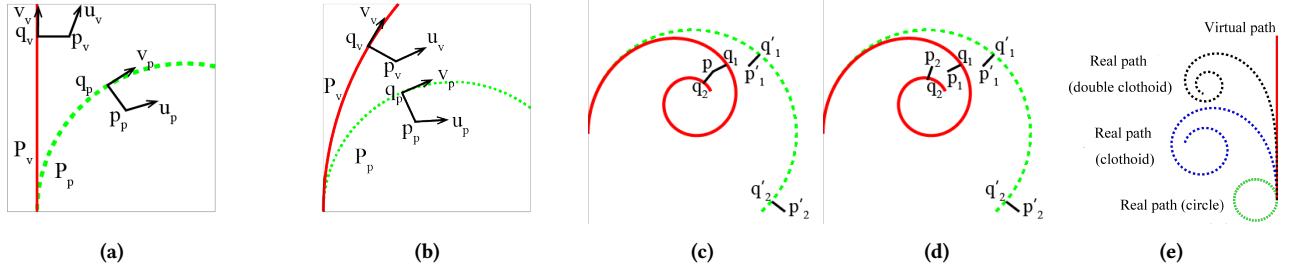


Figure 1: Viewpoint mapping process for dynamic gains: (a) Curvature gain, (b) Bending gain. Errors in dynamic bending gain: (c) Non-unique mapping points in linear bending gain; (d) Mapping errors in linear bending gain. (e) double clothoid curve in quadratic curvature manipulation.

position. Similar to dynamic bending gain, mapping errors in dynamic curvature gain occur during the process of mapping from the physical viewpoint to the virtual viewpoint.

In addition to the mapping errors with dynamic curvature and bending gains, existing viewpoint mapping algorithms have other limitations. These algorithms are typically tailored to specific redirection gains, so when the gain function changes, a new mapping algorithm must be developed. For example, a recent work extends Sakono et al.'s linear dynamic gain to a quadratic dynamic gain using a double clothoid curve (Fig. 1(e)). The formula for the double clothoid is: $X = X_0 + \int_0^s \cos\theta ds$, $Y = Y_0 + \int_0^s \sin\theta ds$, $\theta = c_0 + c_1s + c_2s^2/2 + c_3s^3/6$. Both clothoid and double clothoid curves exhibit spiral characteristics, leading to mapping errors.

We design a viewpoint mapping algorithm based on the unified representation model to ensure the continuity of the virtual viewpoint when the user deviates from the physical path. The proposed method first identifies all perpendicular points on the physical path corresponding to the user's physical viewpoint. The closest point to the user's previous corresponding point is then selected. Using the mapping relationship between the physical and virtual paths, the corresponding point on the virtual path is determined. Finally, the user's virtual viewpoint is computed based on the deviation between the physical viewpoint and its corresponding point. The

Algorithm 1 Viewpoint Mapping Algorithm based on Unified Representation Model

- 1: **Input:** Physical path P_p , virtual path P_v , gain parameters $[C_p, C_v, T, i_p, i_v]$, user's physical viewpoint (p_p, u_p) , and the previous physical viewpoint's corresponding point $P_p(s_{prev})$
 - 2: **Output:** User's virtual viewpoint (p_v, u_v)
 - 3: Based on $(p_p - P_p(s_i)) * v_p(s_i) = 0$ (Equation 5) and $(p_p - P_p(s_i)) * v_p(s_i) = 0$ (Equation 6), find the corresponding point $q_p = P_p(s_p)$ of the user's physical position p_p on the physical path P_p , where s_p is the path length of the closest perpendicular point on P_p to the previous corresponding point $P_p(s_{prev})$. (Fig. 2(a)). Here, if the angle between the vector $P_p(s_{prev})p_p$ and the tangent direction of $P_p(s_{prev})$ is less than 90° , the constraint $s_p > s_{prev}$ is applied in Equation 5; otherwise, the constraint $s_p < s_{prev}$ is applied.
 - 4: Compute the corresponding point on the virtual path: $q_v = P_v(s_v) = P_v(T(s_p))$.
 - 5: Compute the user's virtual position: $p_v = q_v + (p_p - q_p)$.
 - 6: Rotate v_v by the clockwise angle from the tangent direction v_p at q_p on P_p to u_p to get u_v .
-

mapping process from the virtual viewpoint to the physical viewpoint (Fig. 2(b)) is similar to Algorithm 1, with the key difference

being the swapping of the subscripts p and v , and replacing $T(s_p)$ with $T^{-1}(s_v)$ in line 4.

4 Experiment

To evaluate the effectiveness of the proposed viewpoint mapping method, we compared the performance of the existing algorithm with that of Algorithm 1 under different gain conditions. Additionally, we conducted comparative experiments using gain methods randomly generated based on the proposed gain model.

4.1 Experiment Design and Measurement

In real-world walking scenarios, the trajectory of a user often deviates from the theoretical path. To simulate this, we randomly generated motion trajectories under each gain method. A computer algorithm created random walking paths and two viewpoint mapping algorithms were applied to generate the corresponding virtual trajectories. Since existing algorithms can accurately map constant and linear translation gains, our experiments focused on dynamic curvature. For the linear curvature gain, we set $C_p(s) = ac \cdot s$, $C_v(s) = 0$, and $T(s) = s$, with a_c as a randomly generated parameter. We then randomly generated 1000 gain method conditions based on the parameter configurations outlined in Table 1.

Next, we conducted comparative experiments on various gain methods generated from the gain model, including translation, curvature, bending, and gradient gains, as well as their combinations. To do this, we set $C_p(s)$, $C_v(s)$, and $T(s)$ as polynomial functions of degree up to 5, with randomly generated coefficients for each term, generating a total of 3×10^3 conditions. For each gain method, we randomly generated 1000 physical walking trajectories, each consisting of discrete points with equal step lengths. The trajectory generation strategy was as follows: the vector from the previous point to the current point was considered the orientation, and a new point was randomly selected along an arc of a sector centered on this orientation, with the sector's radius corresponding to the step length. Trajectories were grouped according to the maximum distance from the physical path ($d < 0.1$, $0.1 < d < 0.2$ and $d > 0.2$). In total, we collected 5×10^6 data points, including virtual trajectories generated by the existing algorithm and the Algorithm 1.

We evaluated the performance of the algorithm by assessing the continuity of the mapped virtual trajectory. Since the step length of the randomly generated physical trajectory is constant for each step, we measured the continuity of the virtual trajectory by comparing the distance between any two consecutive points $p_v(s_i)$ and $p_v(s_{i+1})$ on the virtual trajectory with a given threshold t . In actual tests, the step length for discrete points on the physical trajectory was 0.02, and t was 0.03. Based on the test data, the maximum distance between adjacent discrete points on a continuous virtual trajectory is 0.024, while the minimum distance between adjacent points on a discontinuous virtual trajectory is 0.72.

4.2 Result

Figs. 3(a), (b) and (c) show results from the linear curvature gain. In Fig. 3(a), the green curve represents the physical trajectory for the linear bending gain with parameters $[1.0 + 0.4s, 0, s, 0, 0]$, while the blue dashed line represents a randomly generated physical trajectory simulating the actual walking of the user. In Figs. 3(b)

and 3(c), the red straight line represents the virtual path, and the yellow curve represents the virtual trajectory computed by the two mapping algorithms. The virtual trajectory of the existing algorithm (orange) is discontinuous, while the one from Algorithm 1 (yellow) is continuous. A similar trend is observed in Figs. 3(d), (e), and (f) for the gain method $[1.0 + 0.4s - 0.1s^2 - 0.01s^3, 0, s, 0, 0]$. Statistical results from 2×10^6 experimental trials show that all virtual trajectories generated by Algorithm 1 are continuous, whereas mapping errors occurred in the trajectories generated by the existing algorithm. The error rates were 31.20% for $d < 0.1$, 54.24% for $0.1 < d < 0.2$, and 100% for $d > 0.2$.

We conducted additional comparisons by generating various gain methods based on our model. Statistical results from 3×10^6 trials show that all virtual trajectories generated by Algorithm 1 are continuous, while trajectories generated by the existing algorithm exhibit mapping errors. These results follow a trend similar to that presented earlier.

5 Discussion

5.1 Analysis of Mapping Errors in the existing algorithm

The errors in the previous viewpoint mapping algorithm [16] arise from the short walking distances used in their experiments. Under these conditions, the physical path had not yet developed the characteristics of a spiral. Our analysis shows that mapping errors in their algorithm arise in regions with multiple nearest points on the path, especially when the user crosses the self-bisector line (Fig. 4). In such cases, the existing algorithm inaccurately maps the user's position, as demonstrated when the user moves from point p_1 to point p_2 after crossing the bisector (Fig. 4(a)). In dynamic curvature gain scenarios, as the Clothoid's curvature increases, the distance between path points and the bisector decreases, causing higher error rates in the existing algorithm as d increases. Longer walking distances further amplify errors due to crossing the bisector. In contrast, our algorithm leverages the user's previous physical viewpoint to ensure accurate mapping, even when the trajectory crosses the bisector (Fig. 4(b)).

5.2 Proof of Mapping Uniqueness of the Proposed Algorithm

As discussed earlier, the existing algorithm suffers from non-unique mappings. In contrast, Algorithm 1 avoids this problem. If it were to result in a non-unique mapping, two perpendicular points $q_1 = P_p(s_1)$ and $q_2 = P_p(s_2)$ would exist on the physical path, where $|s_1 - s_p| = |s_2 - s_p|$ and $s_1 \neq s_2$, meaning they lie on opposite sides of $P_p(s_p)$. However, due to the consideration of the user's movement direction in the algorithm, this situation will not occur unless the user's forward direction aligns with the normal vector of $P_p(s_p)$. In such a case, the physical position p will still correspond to $P_p(s_p)$, preserving the mapping.

5.3 Algorithm Efficiency and Model Flexibility in 3D Environments

In the viewpoint mapping algorithm, calculating the nearest point to a path is simple for constant gain scenarios but becomes complex in

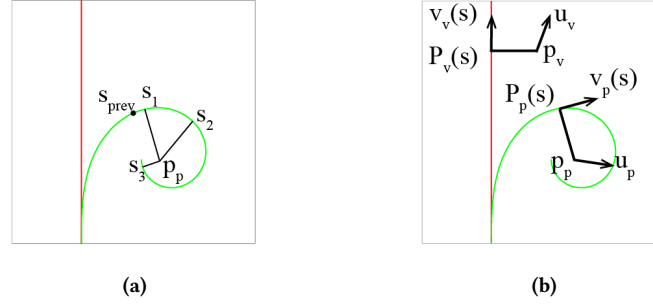


Figure 2: (a) Perpendicular feet s_1 , s_2 , and s_3 of physical position P_p on the physical path. (b) Viewpoint mapping process.

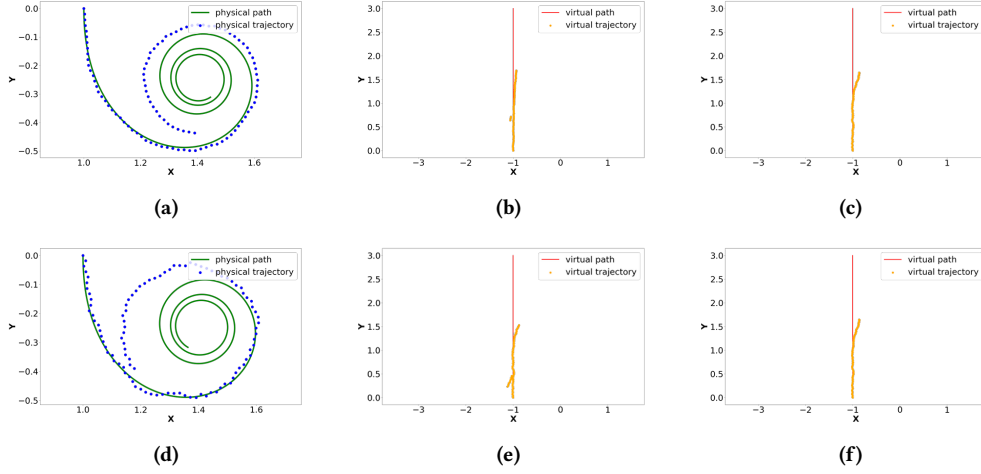


Figure 3: (a) Physical path (green) of the linear curvature gain with the parameters $[1.0+0.4s, 1.0, s, 0, 0]$ and the physical trajectory (blue). (b) Virtual path (red) and virtual trajectory (orange) computed by the previous algorithm. (c) Virtual path (red) and virtual trajectory (orange) computed by our presented algorithm. (d)(e)(f) Another case involves the cubic curvature gain with the parameters $[1.0 + 0.4s - 0.1s^2 - 0.01s^3, 0, s, 0, 0]$.

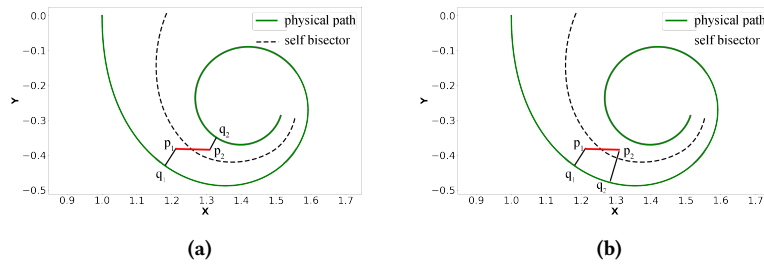


Figure 4: (a) Mapping of the existing algorithm; (b) Mapping of Algorithm 1.

dynamic gain scenarios, such as with Clothoid curves. The existing algorithm discretizes the curve and checks all points, while our approach uses user's previous physical viewpoint to calculate only the local nearest point, significantly reducing computation time. As a result, our algorithm is 13 times faster in dynamic gain scenarios,

achieving the same precision in just 2.29 seconds, compared to the existing algorithm's 29.79 seconds.

Our unified model allows users to easily extend existing gain methods, such as representing constant and linear curvature gains

with specific formulas. By modifying parameters, users can easily transition to higher-order curvature gains or composite gain methods, streamlining the design and implementation process.

In 2D space, after deviating from the path, the user's position is limited to the lateral deviation on either side of the path. In 3D space, however, the user may deviate in any direction along a plane defined by the tangent of the path. While our algorithm accounts for this, validating its performance in a 3D space is challenging due to the complexity of 3D trajectories. For now, we have treated 3D deviations as lateral for validation, but future work will focus on optimizing and validating the algorithm for all directional deviations in 3D.

6 Conclusion

This study addresses the phenomenon of teleportation present in existing redirected walking techniques by proposing a robust viewpoint mapping algorithm specifically designed to accommodate path-mapping-based gains. Through a careful examination of existing gain methods, we developed a gain representational model that supports the design of viewpoint mapping algorithms for various gain strategies. This not only enhances the user's experience of exploring a larger virtual space within a confined physical environment, but also provides new insights and solutions for future augmented and virtual reality applications. Our research lays a solid foundation for the further development of redirected walking technologies and showcases the potential to improve the user experience in dynamic and complex scenarios.

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